

**MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION
Shikshan Prasarak Mandal's**

SPM POLYTECHNIC, KUMATHE, SOLAPUR



MICRO PROJECT

Academic year: 2024-25

TITLE OF PROJECT

Clock divider using FPGA with VHDL Code

Name of the Program :- Electronics and Telecommunication

Program code :- I- Scheme

Name of the Course :- VLSI With VHDL

Course code :- 22062



MAHARASHTRA STATE
BOARD OF TECHNICAL EDUCATION
CERTIFICATE

This is to certify that **Mr. Kokkul Balaj Shrinivas** Roll No. **5 3 4** of **6th** Semester of Diploma in Electronics and Telecommunication Engineering of Institute, **SPM POLYTECHNIC, Kumathe, SOLAPUR** (Code:**0523**) has completed the **Micro Project** satisfactorily in Subject VLSI With VHDL (Code:22062) for the academic year **2024 - 2025** as prescribed in the curriculum.

Place : Kumathe

Enrollment No : 2321140019123211400191

Date : / / 2025

Exam Seat No. :

(Mr. Choudhari A.V.)
Course Coordinator

(Mr. Choudhari A.V.)
Head of the Program

(Mrs. Rohini Chavan)
Principal

Annexure – IA

Format for Micro-Project Proposal

Semester: - Sixth

Course:-VLSI With VHDL

Title of Micro – Project:- Clock divider using FPGA with VHDL Code

1.0 Brief Introduction- (Importance of the project):-

Clock Divider is also known as frequency divider, which divides the input clock frequency and produce output clock. In our case let us take input frequency as 50MHz and divide the clock frequency to generate 1KHz output signal. VHDL code consist of Clock and Reset input, divided clock as output. A clock divider circuit creates lower frequency clock signals from an input clock source.

2.0 Aim of the Micro – Project: -

The aim of this project is to implement a clock divider using VHDL code on an FPGA, which reduces the frequency of an input clock to a desired lower frequency, enabling the design to operate at a different, slower rate.

3.0 Action Plan (Sequence and time required for major activities for 7 Weeks)

Sr. No.	Details of Activity	Planned Start date	Planned Finish date	Roll no of Responsible Team Member
1	Discussion of topic			505,517,534
2	Finalization of topic			505,517,534
3	Collection and finalization of Information			505,517,534
4	Debug The Program			505,517,534
5	Error Correction			505,517,534
6	Prepare Report			505,517,534
7	Submission of Report			505,517,534

ANEEXURE II

Evaluation Sheet for the Micro Project

Academic Year : 2024-25

Name of the Faculty: Mr. Choudhari A. V.

Course: VWV

Course code : 22062

Semester :EJ-6I

Title of the project : Clock divider using FPGA with VHDL Code

COs addressed by Micro Project : Maintain FPGA based circuit

Major learning outcomes achieved by students by doing the project

(a) Unit outcomes in Cognitive domain:-Describe VHDL simulation for the given application

(b) Outcomes in Affective domain:

1) Function as a Team-Member.

2) Follow the Ethics.

Comments/suggestions about team work /leadership/inter-personal communication (if any)

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Roll No	Student Name	Marks out of 6 for performance in group activity (D5 Col.8)	Marks out of 4 for performance in oral/ presentation (D5 Col.9)	Total out of 10
505	Kokkul Balaj Shriniwas			

Mr. Choudhari A.V.
Course Coordinator

WEEKLY PROGRESS REPORT

MICRO PROJECT

Sr. No.	Details of Activity	Planned Start date	Planned Finish date	Roll no of Responsible Team Member
1	Discussion of topic			505,517,534
2	Finalization of topic			505,517,534
3	Collection and finalization of Information			505,517,534
4	Debug The Program			505,517,534
5	Error Correction			505,517,534
5	Prepare Report			505,517,534
6	Submission of Report			505,517,534

(Kokkul Balaj Shriniwas)

Name of Student

Mr. Choudhari A.V

Course Coordinator

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1. Introduction

Clock division is a fundamental operation in FPGA-based designs, where a high-frequency clock signal is divided down to a lower frequency. This is useful for timing control, reducing power consumption, and interfacing with slower peripherals.

A clock divider takes an input clock signal and produces an output clock signal with a lower frequency. The division factor determines how many input clock cycles are needed for one output clock cycle. This is typically achieved using counters or flip-flops.

A clock divider is a crucial component in FPGA-based designs used to generate lower-frequency clock signals from a high-frequency input clock. This is essential for interfacing with peripherals, reducing power consumption, and managing timing constraints.

2. Working

A clock divider is a circuit that reduces the frequency of an input clock signal by a specific factor. In FPGA-based designs, it is implemented using counters or flip-flops.

A clock divider reduces the frequency of an input clock signal by a specified factor. Since FPGAs operate using high-speed system clocks, a clock divider is necessary to generate lower-frequency clocks for different functional blocks, peripherals, or synchronization.

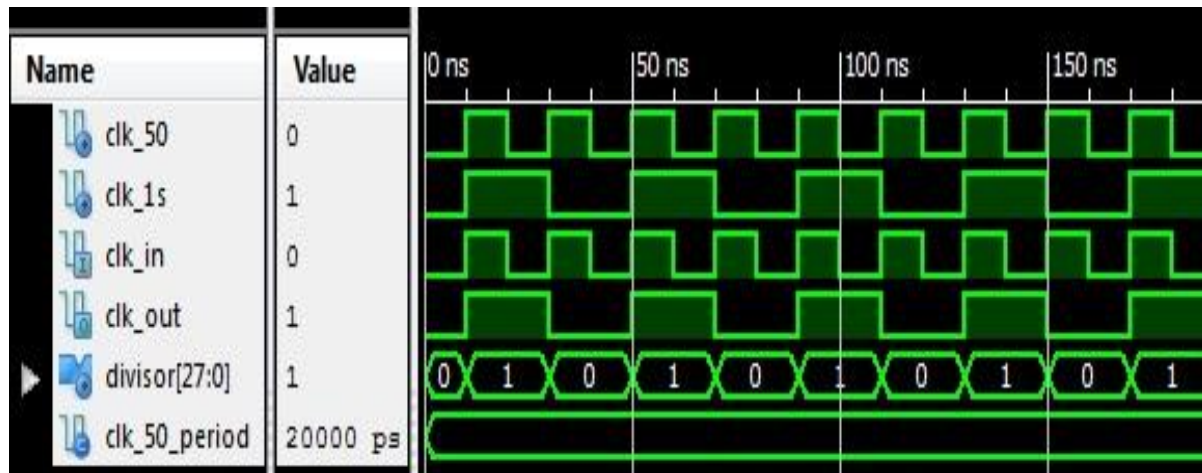
1. **Input Clock Signal:** The FPGA receives a high-frequency clock (clk_in).
2. **Counter Mechanism:** A counter increments on every clock pulse.
3. **Toggle Output:** When the counter reaches a predefined value, the output clock (clk_out) toggles, effectively reducing the frequency.
4. **Reset Mechanism:** A reset signal ensures that the clock divider starts from a known state.

3. Code

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
USE IEEE.STD_LOGIC_UNSIGNED.ALL;
entity clock_div is
port (
clk_in: in std_logic; -- clock input on FPGA
clk_out: out std_logic -- clock output
);
end clock_div;

architecture Behavioral of clock_div is
signal divisor: std_logic_vector(27 downto 0) := (others => '0');
begin
process(clk_in)
begin
if(rising_edge(clk_in)) then
divisor <= divisor + x"00000001";
-- If(divisor>=x"2FAF07F") then -- for running on FPGA -- comment when running simulation
-- Modify the divisor (x"2FAF07F"=499999999) above to get the clock frequency you want:
-- Frequency of clk_out = Frequency of (clk_in) divided by (divisor + 1)
-- If the frequency of clk_in is 50MHz and the divisor is 499999999=x"2FAF07F",
-- the frequency of clk_out is 1Hz
if(divisor>=x"00000001") then -- for running simulation -- comment when running on FPGA
-- divisor = 1 => divide clock by half for simulation purposes
divisor <= x"00000000";
end if;
if(divisor<x"00000001") then -- replace x"00000001" by x"17D7840" when running on FPGA
clk_out <= '1';
else
clk_out <= '0';
end if;
end if;
end process;
end Behavioral;
```


4. Output



5. Advantages

1. Portability
2. Flexibility
3. Run-time Reconfiguration
4. Avoidance of Dedicated Resources
5. Timing Performance

6. Applications

1. Generating Slower Clocks
2. Timing Signals
3. Synchronizing Different Clock Domains
4. Counter Implementation

7. Conclusion

Clock dividers are crucial for FPGA-based designs, enabling frequency scaling, power efficiency, and better synchronization. They are simple to implement using VHDL and use minimal FPGA resources.

A clock divider is an essential component in FPGA-based designs, allowing the generation of lower-frequency clock signals from a high-frequency input clock. It is widely used for timing control, synchronization, power efficiency, and peripheral interfacing.

In this implementation, a counter-based approach was used to toggle the output clock after a specific number of input clock cycles. The VHDL code effectively demonstrates how to divide the clock frequency using FPGA resources efficiently.

8. Reference

1. Peter J. Ashenden, *The Designer's Guide to VHDL*, Morgan Kaufmann, 2010.
2. Pong P. Chu, *FPGA Prototyping by VHDL Examples: Xilinx Spartan-3 Version*, Wiley, 2008.
3. Craig Hackney, "PGA Motor Control Reference Design," Application Note: Spartan and Virtex FPGA Families, Xilinx XAPP808 vol. 1.0, September 16, 2005.
4. Anthony Cataldo, "Low-priced FPGA options set to expand," *Electronic Engineering Times Journal*, no. 1361, pp. 38-45, USA, 2005.
5. Gordon Hands, "Optimised FPGAs vs dedicated DSPs," *Electronic Product Design Journal*, vol. 25, no. 12, UK, December 2004.
6. Volnei A. Pedroni, *Circuit Design with VHDL*, MIT Press, 2004.